

ML module #5

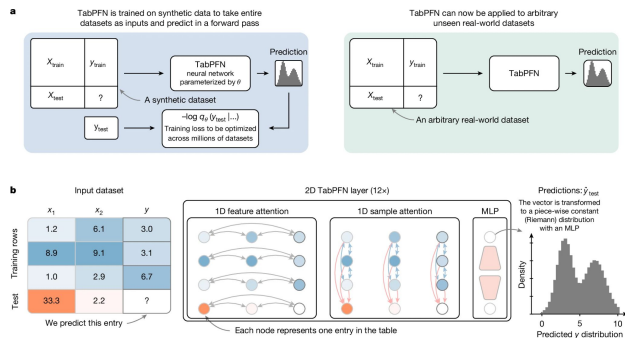
(TabPFN)

Benjamin Sanchez-Lengeling

CrossTALK: Cross-Training in AI and Laboratory Knowledge for
Drug Discovery.

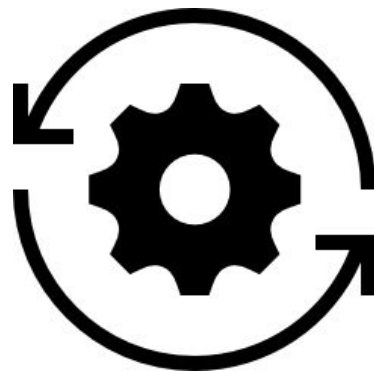


Panorama



Self-Attention

Pre-training



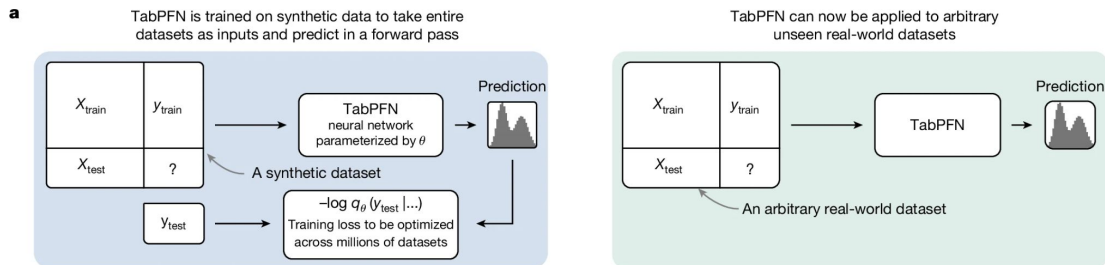
TabPFN

Working sesh

Activity: Talk to someone spatially distant

1. What ideas do you have for modelling?
2. How do we pick a metric to evaluate ML models for drug discovery?
3. What tradeoff do we have to think about when proposing candidate molecules?

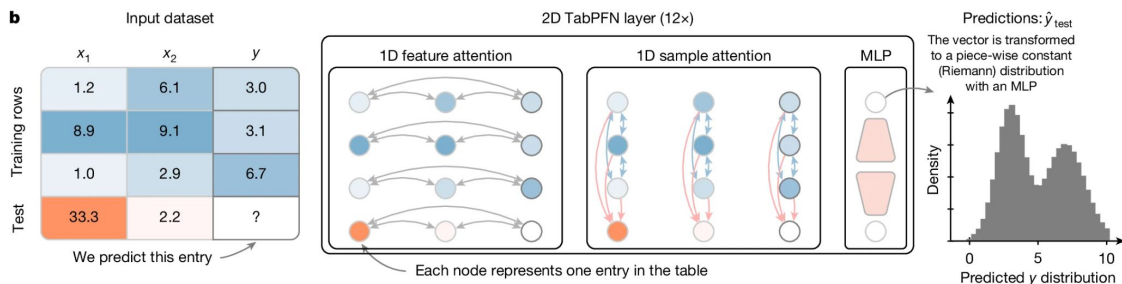
TabPFN: Foundation Model for Tabular Data



Important to note:

b) Self-attention

a) Pre-training



<https://github.com/PriorLabs/TabPFN>

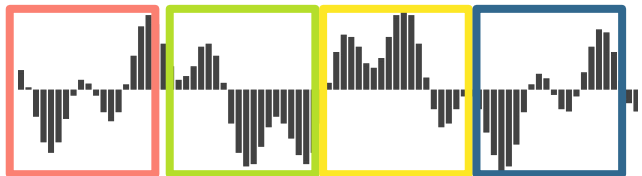
Tokens: Inputs for Transformer/Attention layers

Numerical representations for pieces of data.

Text: Text chunks to embeddings

T o m a t o e s a r e o n e o f t h e m o s t p o p u l a r p l a n t s f o r v e g e t a b l e g a r d e n s .

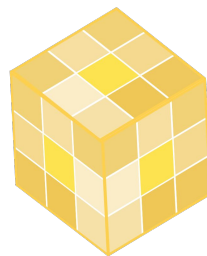
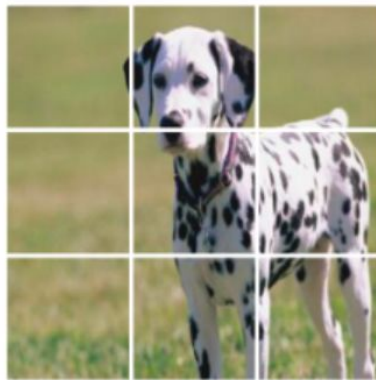
Audio? Sound wave chunks



Graphs? Nodes are tokens



Images: Patches



[Batch, Seq Length, Dim]
3-Tensors

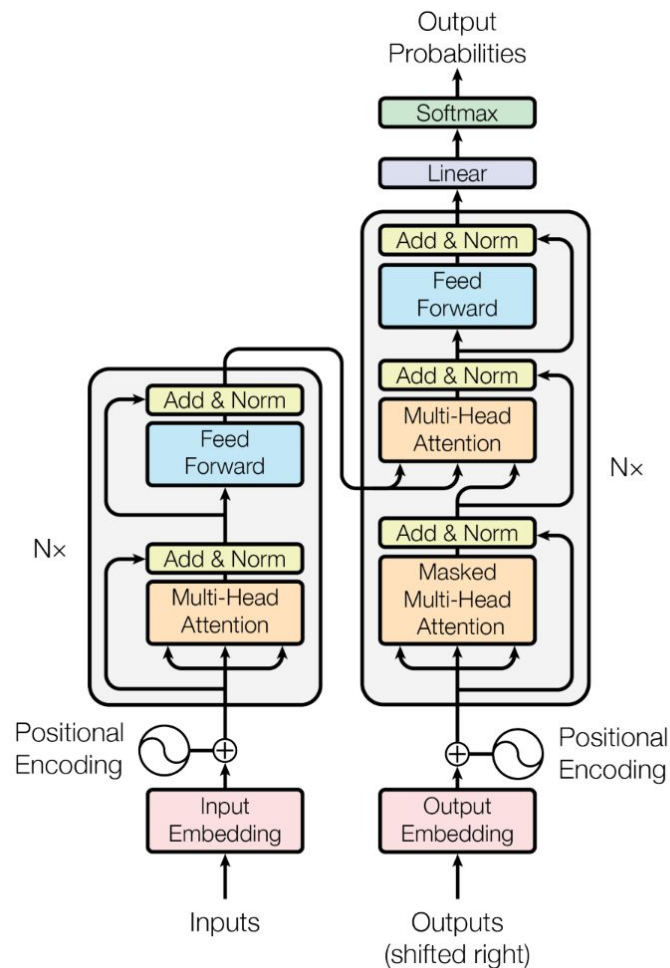
Transformers in a Nutshell

Layers of:

- Attention Mechanism (Multi-headed)
- Positional Encodings

With massive unsupervised datasets:

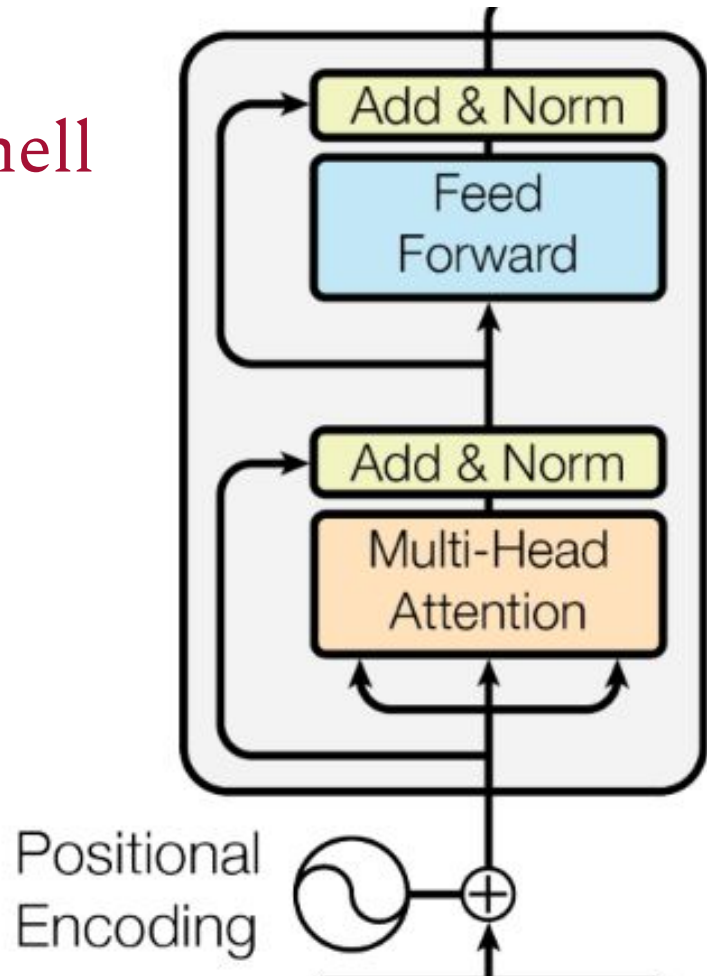
- Masked self-supervised training
- Contrastive training



Transformers Layers in a Nutshell

Inside a transformer layer:

- Self-Attention (multi-headed)
- Residual connections
- Layer Norm
- FFN / MLP
- Positional Encodings



What is Attention? A Bilinear Lens

Attention learns relationships dynamically via “learned” inner products.

$$L(x) = Wx$$

$$B(x, y) = x^T W y$$

Bilinear component, n^2 entries

$$\text{score}_{ij} = (q_i \cdot k_j) / \sqrt{d_k}$$

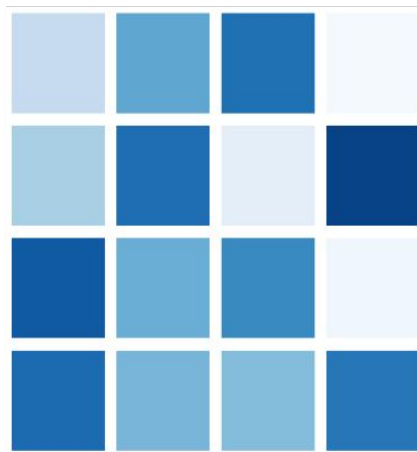
$$B(x, x) = x^T W x$$

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

Attention Maps: Visualizing Where the Model Attends

Understanding the Focus of Attention Mechanisms

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

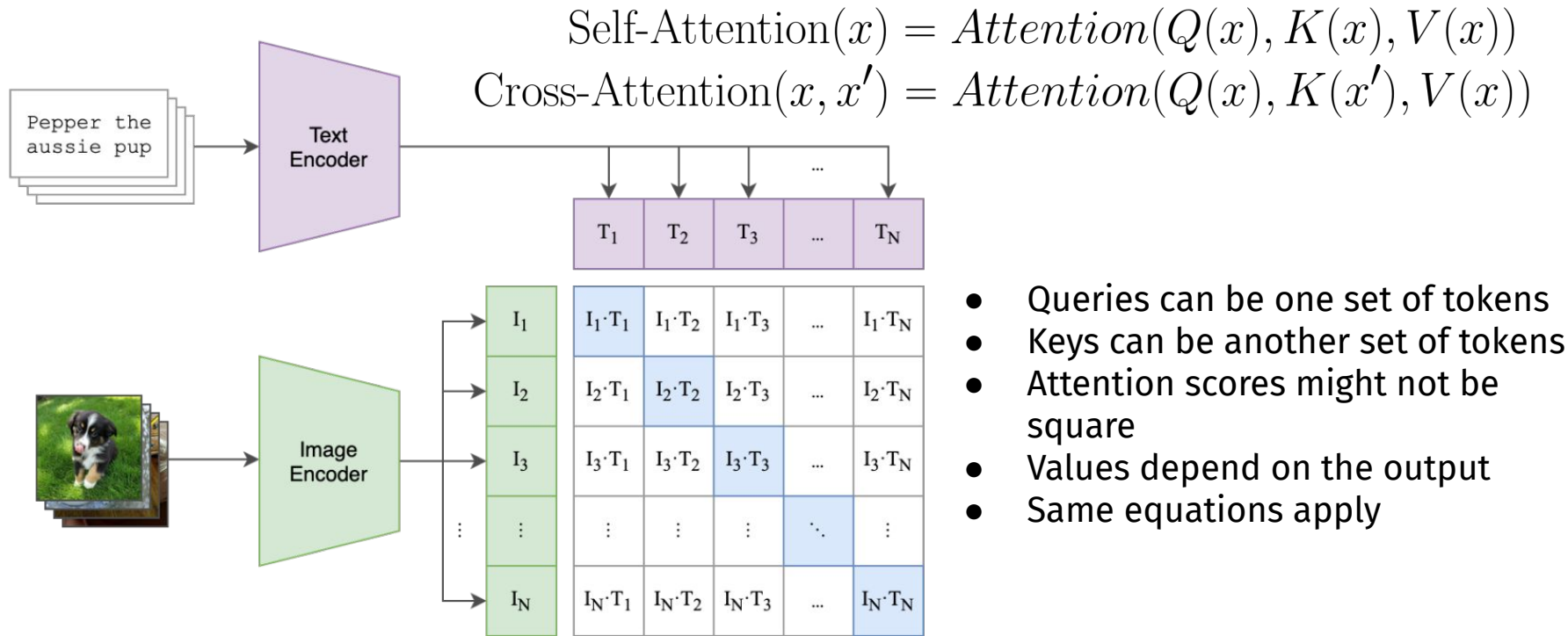


Softmax bias:

- Values between 0 and 1
- Categorical like.
- Attend to one token at a time.

When is Self-Attention not Self? Cross-attention

Applying Attention Between Distinct Inputs

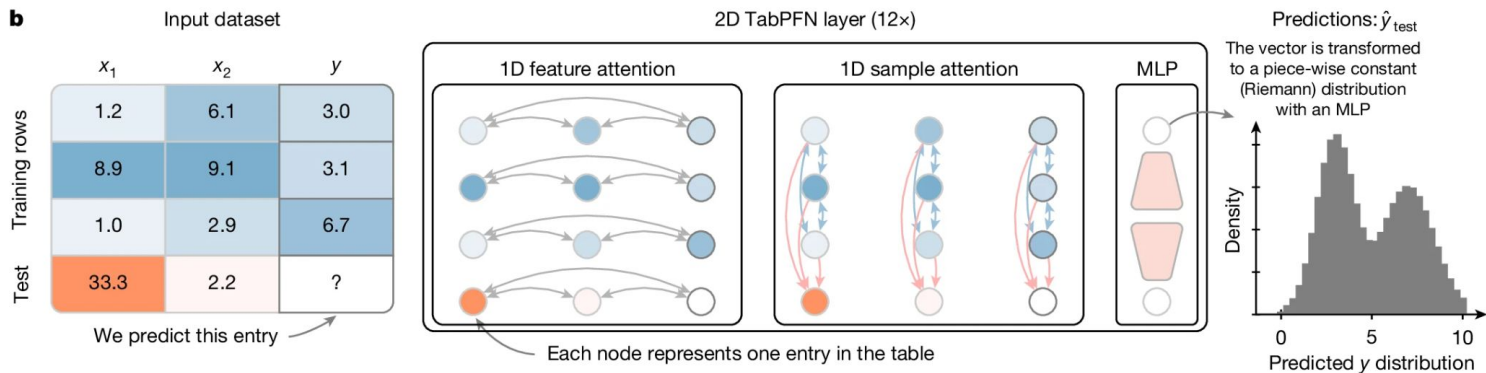


Self-Attention recap

$$\text{Attention}(Q, K, V, M) = \text{softmax} \left(\frac{QK^T M}{\sqrt{d_k}} \right) V$$

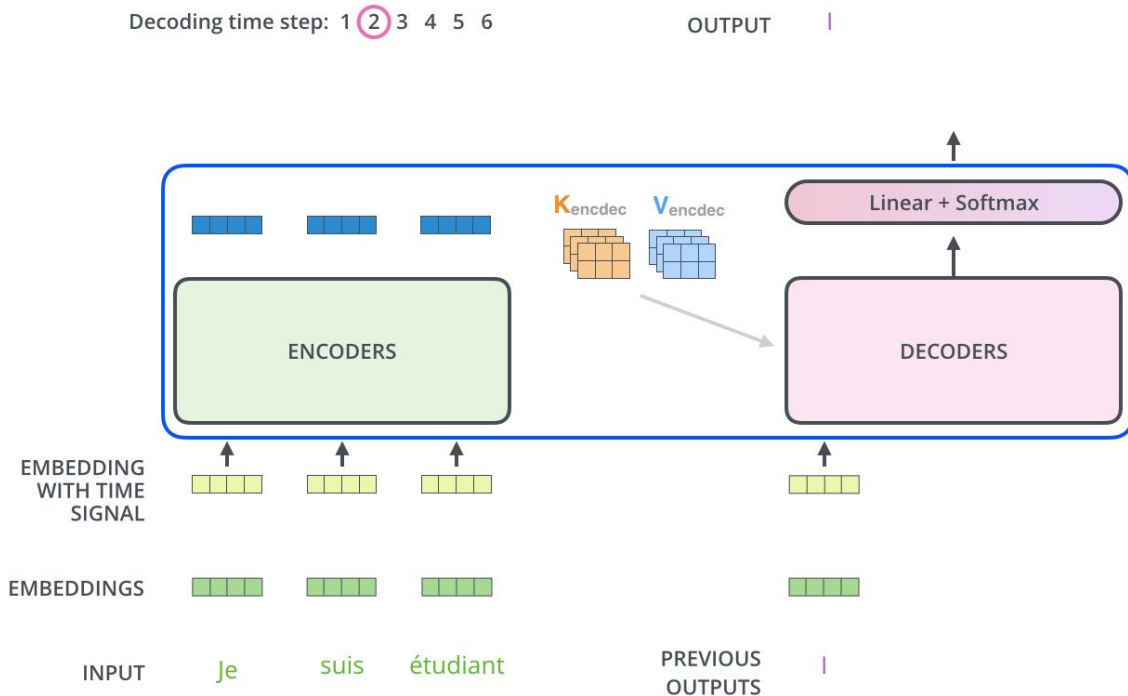
$$\text{Self-Attention}(x, \text{mask}) = \text{Attention}(\text{Linear}(x), \text{Linear}(x), \text{Linear}(x), \text{mask})$$

$$\text{Cross-Attention}(x, x', \text{mask}) = \text{Attention}(\text{Linear}(x), \text{Linear}(x'), \text{Linear}(x), \text{mask})$$



Outputs: Auto-Regressive Decoding of Tokens

Decoding one token at a time, using previous outputs.



Pre-training: Masked Language Modelling (Self-supervised)

Span denoising in T5x as an example

Original text

Thank you ~~for inviting~~ me to your party ~~last~~ week.

Inputs

Thank you <X> me to your party <Y> week.

Targets

<X> for inviting <Y> last <Z>

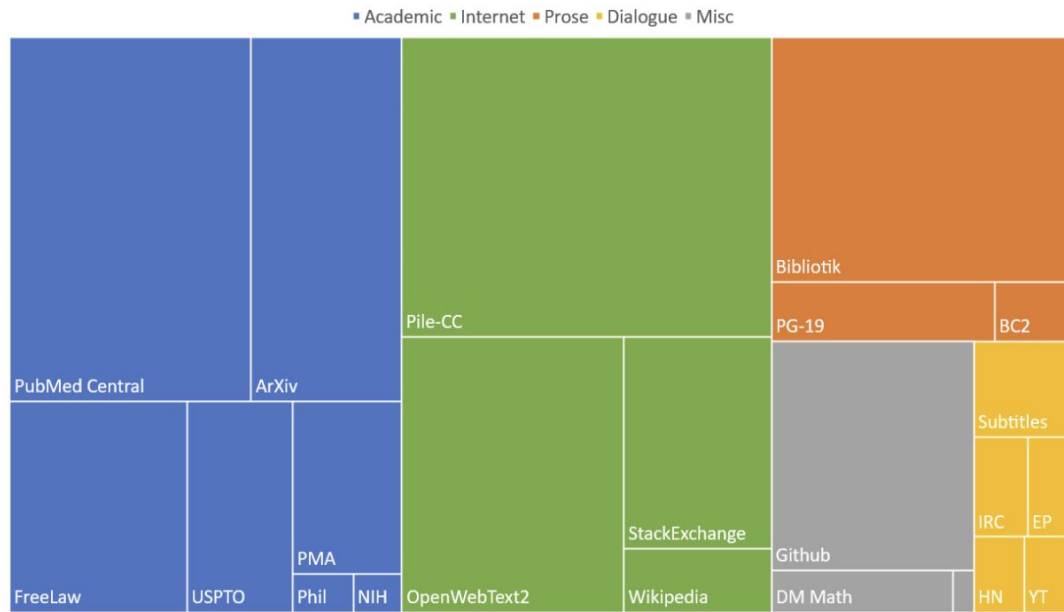
Sizes of Text Datasets for LLMs

Scale significantly impacts model ability.

Common Crawl	225 B web pages	275 TBs
The Pile	22 datasets	825 GiB



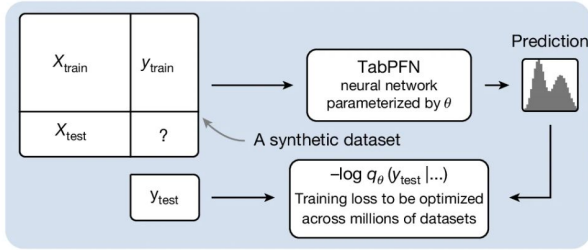
Composition of the Pile by Category



TabPFN's Pre-Training: From Solving One Problem to Learning How to Solve *Any* Problem

a

TabPFN is trained on synthetic data to take entire datasets as inputs and predict in a forward pass



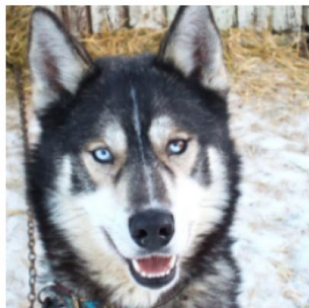
A **single, powerful model** is pre-trained once on millions of *simulated* datasets. It learns a universal algorithm for finding patterns in tabular data.

The core mechanism is **in-context learning**. At inference time, TabPFN doesn't update its weights; it uses your training data as the instruction manual for how to make the next prediction.

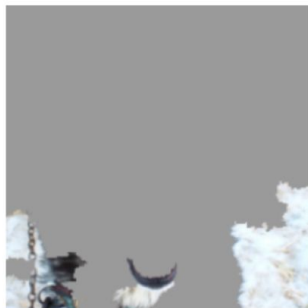
- For a Large Language Model (LLM), the "**Prompt**" is a string of text.
- For TabPFN, the "**Prompt**" is your entire training dataset (X_{train}, y_{train}).

Spurious correlations and dataset biases

Statistical patterns in our dataset can affect the weights (and explanations) of our model



(a) Husky classified as wolf



(b) Explanation

Figure 11: Raw data and explanation of a bad model's prediction in the "Husky vs Wolf" task.

	Before	After
Trusted the bad model	10 out of 27	3 out of 27
Snow as a potential feature	12 out of 27	25 out of 27

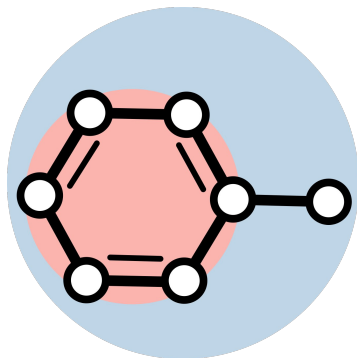
Table 2: "Husky vs Wolf" experiment results.

What other kinds of biases might we find IRLab?

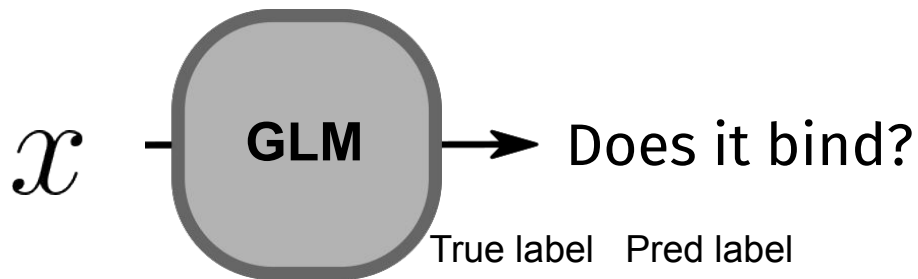
Causal vs Spurious Correlation in Molecules

"Causal relationship":

Binding is due to benzene ring



"Spurious Correlation":
Co-occurs with the
real mechanism



Real mechanism:
Binding is due to
benzene ring

	✓	✓
	✓	✓
	✓	✓
	X	X

	✓	X / ✓
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Training Set

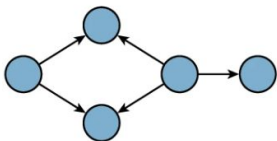
Test Set

?

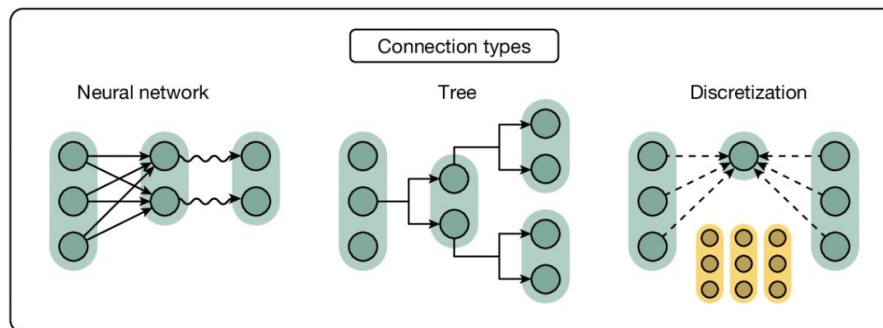
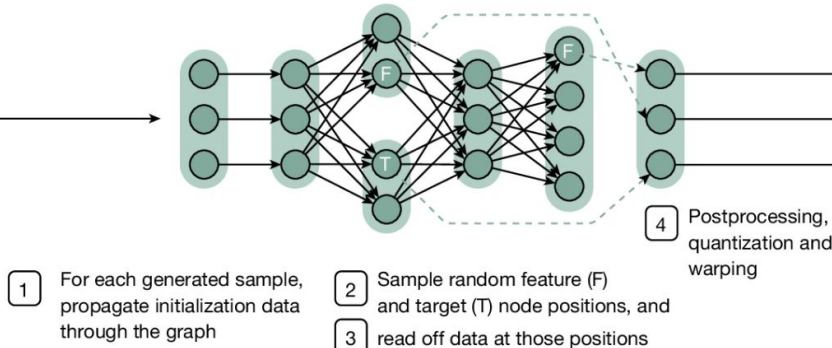
Synthetic dataset generation via Structured Causal Models

a Sample underlying parameters

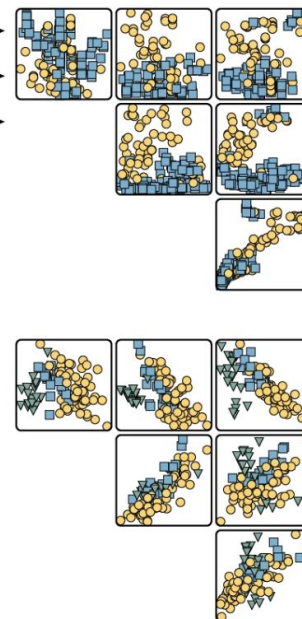
Sample number of data points
Sample number of features
Sample number of nodes
Sample graph complexity
Sample graph



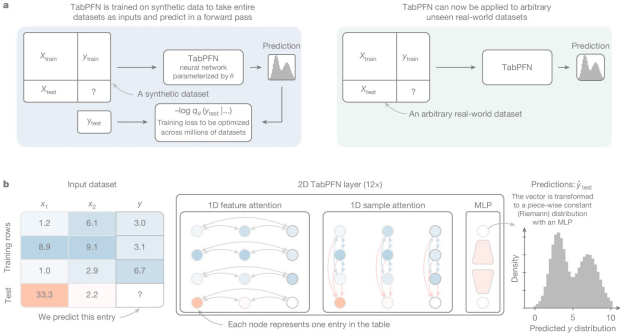
b Build computational graph and graph structure



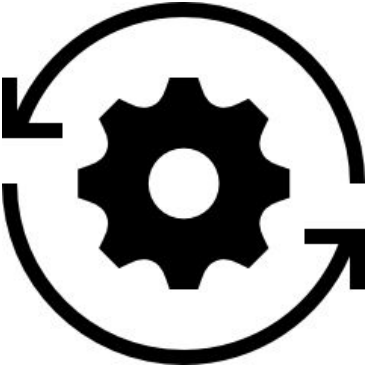
c Final datasets



Panorama



TabPFN



Working sesh

Results...
Drumrolls

Leaderboard

#	Team	Members	Score	Entries	Last	Join
1	NoMore9AMClassesPlease	 	7.00000	29	2d	
2	DELtheASMS!	  	7.00000	30	8h	
3	DELulu is the Solulu		7.00000	7	5d	
4	PON-Team	  	4.00000	8	2d	
5	Crème DEL la overfit!!!	  	2.00000	11	9h	

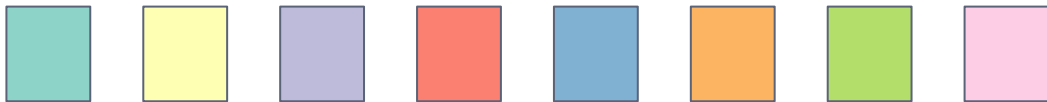
For color palette

<https://colorbrewer2.org/#type=sequential&scheme=BuGn&n=3>

Dark2



Set3



Cividis (continuous)



PiYG (divergent)

